

Edward (Ned) Pearson

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A recent Bachelor of Computer Science graduate from RMIT with a strong technical foundation across software development, data science, and machine learning. Experienced in delivering projects end to end, from understanding a problem and defining requirements through to building, evaluating, and communicating practical solutions. Comfortable working both independently and in teams, engaging directly with clients and stakeholders, and communicating technical work clearly to technical and non-technical audiences alike. Committed to careful, honest work and to continuously growing my skills.

EDUCATION

Bachelor of Computer Science

RMIT | 2021 – 2026

Completed a broad computer science degree spanning software development, data science, machine learning, and cloud computing, with a strong applied and project-based focus.

Selected coursework

- **Machine Learning:** supervised and unsupervised techniques, model evaluation, and deep neural networks applied to real-world datasets using open-source toolkits; built complete end-to-end ML systems.
- **Practical Data Science:** transforming raw data (text, spatial, and time-series) into analysis-ready form through the full pipeline of data wrangling, modelling, and presentation of results in Python.
- **Artificial Intelligence:** core AI techniques for search, reasoning, and decision-making, and how machines analyse patterns to make predictions.
- **Full Stack Development:** industry-standard web applications with a ReactJS front-end, a Node.js/Express middle layer, and a cloud-hosted MySQL database.
- **Cloud Computing:** cloud architectures, platforms, and services, and designing scalable, distributed applications.
- **Algorithms and Analysis:** algorithm design paradigms and computational complexity analysis.
- **Software Engineering: Process and Tools:** agile development processes, version control, and collaborative engineering practices.

Additional coursework: Operating Systems Principles, Object-Oriented Software Design, Advanced Programming Techniques, Computing Theory, Professional Computing Practice, Software Engineering Fundamentals, Java and C++ Programming Bootcamps and Studios, Mathematics for Computing, and Practical Statistics.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript, SQL
- **Data & Machine Learning:** pandas, NumPy, scikit-learn, PyTorch, exploratory data analysis, model evaluation, data augmentation, CNNs, transfer learning
- **Databases & Data Handling:** SQL, MySQL, Firestore (NoSQL), data validation & quality checks, web scraping, AI API integration
- **Visualisation:** Matplotlib, seaborn, dashboards
- **Web & Full Stack:** React, Node.js, Express, HTML, CSS, REST APIs
- **Cloud, Tools & Practices:** AWS, Docker, Git, Agile/Scrum

UNIVERSITY PROJECTS

Capstone: Your Accessible Flight (Accessibility Data Automation)

Team of 5

- Worked directly with an external client, an aviation-accessibility start-up, to define requirements and deliver a system that automated a previously manual data workflow spanning dozens of airlines and airports.
- Built in Python, the pipeline extracted data from web sources and ingested PDFs, used an AI API to structure it into consistent records, and validated the output through automated quality checks; change detection ensured existing data was updated only where it had genuinely changed, keeping the dataset accurate and protecting clean records.
- Built a dashboard presenting each pipeline stage visually, with extraction results filterable by category, validation pass rates, and version differences, so the state of the data was clear at a glance.
- Consulted both non-technical and technical client stakeholders through weekly reviews, tailoring communication to each; ran delivery in agile sprints with a shared backlog and Git version control.
- Delivered a fully working, documented system that was demonstrated, accepted, and has since replaced the client's manual process.

Colorectal Cancer Cell Classification

Individual: Python, scikit-learn, PyTorch

- Extracted and analysed a histopathology image dataset, then built and compared five approaches (logistic regression, SVM, CNN, CIFAR-10 transfer learning, and a majority-vote ensemble) across binary and four-class classification tasks.
- Designed a patient-level evaluation framework to prevent data leakage, tuned hyperparameters systematically, and applied data augmentation to reduce overfitting by 93%; the augmented CNN met the project's primary performance target.
- Analysed each model's limitations, weighed errors against real-world clinical cost, and reported results that fell short of target rather than only best-case figures, reflecting a commitment to rigorous, honest analysis.

Wildfire Intensity Prediction

Individual: Python, scikit-learn

- Built and compared Decision Tree, SVM, and neural network classifiers to predict wildfire intensity from historical data, applying feature engineering, stratified evaluation, and per-class performance analysis.

Airbnb Price Prediction

Individual: Python, scikit-learn

- Developed a price-prediction model spanning the full pipeline from data cleaning and feature engineering through to model evaluation.

Full-Stack Web Application

React, Node.js/Express, MySQL

- Designed and built a full-stack web application with a React front-end, a Node.js/Express middle layer, and a cloud-hosted MySQL database, following industry-standard development practices.

Further projects are available on my personal website.

WORK EXPERIENCE

Bar Attendant (Casual)

Ascot Vale Hotel | Mar 2025 – Current

- Provide efficient, friendly customer service in a busy, fast-paced hospitality environment.

Assistant Bar Manager (Casual)

Point Lonsdale Surf Life Saving Club | Nov 2022 – Jan 2024

- Managed a team in a fast-paced environment, supporting collaboration and time management.
- Led training for new staff and handled day-to-day administrative tasks.
- Delivered high-quality customer service, balancing customer expectations and business goals.

Administrative Support Worker (Casual)

Nextt Disability Care / NDIS | 2019 – 2020

- Delivered tailored technology training to enhance client independence.
- Provided administrative support and developed programs to meet specific client needs.

Various other casual roles including swimming teacher and food service.

VOLUNTEER EXPERIENCE

Trainer & Assessor / Lifesaver (Volunteer, Seasonal)

Point Lonsdale SLSC | 2016 – Current

- Active member for over ten years, contributing across both patrolling and training.
- As a qualified Trainer and Assessor (held for six years), help deliver the Bronze Medallion course each summer, teaching and assessing new members from a wide range of backgrounds and preparing them to patrol safely.
- Plan and run structured training sessions largely under my own direction, taking responsibility for candidates through to assessment, experience directly relevant to building capability in others.
- Patrol Point Lonsdale Surf Beach as a regular member, working within a team to ensure public safety, often in demanding conditions; also officiate the club's IRB racing team each season.

CERTIFICATIONS

- Surf Life Saving Bronze Medallion
- Trainer and Assessor / Training Officer Certificate (LSV)
- First Aid Certification
- Responsible Service of Alcohol (RSA)
- Victorian Working with Children Check

ELIGIBILITY

- Australian citizen; eligible to obtain and maintain a security clearance.